

Daily Picks Pipeline — Technical Documentation

System Overview

The Daily Picks Pipeline is an automated intraday trading system for the Indian stock market (NSE). It runs at **8:40 AM IST** each trading day — before the market opens at 9:15 AM — and produces a shortlist of high-conviction trade setups with precise entry, stop-loss, and target levels.

The pipeline is designed around a **fail-closed** philosophy: if critical data sources (regime, SGX Nifty, global intel) are unavailable, the system halts rather than trading blind.

Key files:

- `dailyPicksService.js` — Pipeline orchestrator (Steps 1–8)
- `dailyPicksScans.js` — Scan definitions and regime mapping
- `dailyPicksConstants.js` — Shared constants (score thresholds, intel adjustments)
- `earningsFilter.js` — NSE corporate event filter
- `globalMarketIntel.js` — AI-powered global market intelligence + SGX Nifty scraper
- `scanLevels.js` — Levels engine (entry, stop, target, R:R calculation)
- `regime.js` — Nifty structure regime detection

Pipeline Flow Summary

Step 1:	Regime + SGX	-> CONFLICT halts
Step 2:	ChartInk scans	-> regime-gated, NEUTRAL dedup
Step 2.5:	Earnings filter	-> NSE board meetings, 4-day lookahead
Step 3:	Enrich	-> Upstox OHLCV + indicators
Step 4:	Score	-> composite + scan priority bonus
Step 5:	Levels engine	-> entry/stop/target/R:R, ATR-based stops
Step 5.5:	Global intel	-> AI web search, STAY_OUT gate, sector/stock adjustments
Step 6:	Select picks	-> sector cap -> scan diversity -> fill by score
Step 6b:	AI insights	-> 1-2 sentence technical rationale (non-fatal)
Step 7:	Save to DB	
Step 8:	Notification	

Step 1: Market Context (Regime + SGX Nifty)

What it does

Two data sources are fetched **in parallel**. Both are **fail-closed** — if either fails, the pipeline throws an error and sends a notification. There is no fallback.

- Nifty Structure Regime** — Fetches Nifty's previous close and 50-day EMA from `fetchAndCheckRegime()`. Determines whether the market structure is bullish (Nifty > EMA50), bearish (Nifty < EMA50), or flat.
- SGX Nifty (GIFT Nifty)** — Scraped from `sgxnifty.org`. Provides pre-market sentiment via the change percentage. Determines whether overnight sentiment is bullish (>+0.3%), bearish (<-0.3%), or flat.

Combined Regime Decision Matrix

`getCombinedRegime(niftyClose, ema50, giftNiftyChangePct)` merges structure and sentiment into a **7-branch decision tree**:

Structure (Nifty vs EMA50)	SGX Sentiment	Combined Regime	maxTrades	sizeMultiplier
Bull (>0.3% above EMA50)	Bull (>+0.3%)	STRONG_BULL	3	1.0
Bull	Bear (<-0.3%)	WEAK_BULL	2	0.6
Bull	Flat (-0.3~+0.3%)	WEAK_BULL	2	0.6
Bear (<0.3% below EMA50)	Bear	STRONG_BEAR	3	1.0
Bear	Bull	CONFLICT	0	0.0
Bear	Flat	WEAK_BEAR	2	0.6
Flat	Any	NEUTRAL	1	0.5

CONFLICT Halt

CONFLICT (structure bearish + SGX bullish) means contradictory signals. The pipeline **halts immediately** — saves an empty document, sends a notification ("mixed signals, sitting out"), and exits. No scans are run.

Output

`marketContext` object containing: `regime`, `structure_regime`, `nifty_prev_close`, `distance_pct`, `ema50`, `sgx_data`, `size_multiplier`, `max_trades`, `decided_at`.

Step 2: Run Scans (ChartInk)

What it does

Runs ChartInk technical scans based on the regime. Each regime only allows specific scan types — this is the **first filter** that prevents running bearish scans in a bull market and vice versa.

Scan Types (16 Total)

Bullish Momentum (4): `breakout_setup`, `fiftyTwoWeek_high`, `bull_flag`, `volume_shocker_bullish`

Bullish Compression (4): `pullback_at_support`, `compression_bullish`, `nr7_bullish`, `inside_day_bullish`

Bearish Momentum (4): `breakdown_setup`, `fiftyTwoWeek_low`, `bear_flag`, `volume_shocker_bearish`

Bearish Compression (4): `failed_at_resistance`, `compression_bearish`, `nr7_bearish`, `inside_day_bearish`

Regime-Gated Scan Selection

Regime	Scans Allowed	Count	Priority Order
STRONG_BULL	All bullish	8	Momentum-first: breakout, 52W, flag, vol shocker, then compression
WEAK_BULL	Compression	4	pullback, compression, NR7, inside day
NEUTRAL	Low-vol only	4	NR7 bull/bear, inside day bull/bear
WEAK_BEAR	Compression	4	failed at resistance, compression, NR7, inside day
STRONG_BEAR	All bearish	8	Momentum-first: breakdown, 52W, flag, vol shocker, then compression
CONFLICT	None	0	Halted at Step 1

NEUTRAL Cross-Direction Dedup

In NEUTRAL regime, stocks that appear in **both** bullish and bearish scans are removed entirely. If a stock triggers NR7 bullish AND NR7 bearish, it means there's no directional conviction — the stock is just compressed, not directional. These are removed before proceeding.

Exit Condition

Zero candidates after scanning -> saves empty doc, sends notification, exits.

Step 2.5: Earnings / Corporate Event Filter

What it does

Removes stocks with upcoming board meetings or corporate events that could cause unpredictable gaps, invalidating technical setups. Runs **before** enrichment (Step 3) to avoid wasting Upstox API calls on stocks that will be discarded.

Data Source

NSE Board Meetings API: <https://www.nseindia.com/api/corporate-board-meetings?index=equities>

How it works

1. Fetches the full list of upcoming board meetings from NSE
2. For each meeting, checks both `bm_purpose` (structured: "Financial Results", "Dividend") and `bm_desc` (free-text detail) for risky keywords
3. Keywords: `result`, `financial`, `quarterly`, `annual`, `dividend`, `buyback`, `fund rais`
4. Meeting date must be within **4 days** of today (lookahead window)
5. All date comparisons are **IST-aware** (the server may run in UTC, but NSE dates are Indian market dates)

Fail-Open Design

If the NSE API is unreachable (timeout, HTTP error), **all candidates pass through**. This is intentional — a network glitch shouldn't block the entire pipeline. The earnings data is supplementary, not critical.

Cache

4-hour TTL, keyed by IST date. First call each day fetches fresh data; subsequent calls within the same day use cache.

Exit Condition

All candidates removed by earnings filter -> saves empty doc, exits.

Step 3: Enrich with OHLCV + Indicators

What it does

Calls the **Upstox API** for each surviving candidate to fetch real market data:

- **OHLCV:** Open, High, Low, Close, Previous Close, Volume
- **Indicators:** EMA20, ATR (Average True Range), RSI (Relative Strength Index)
- **Additional:** Weekly close, weekly EMA20 (for multi-timeframe confirmation)

Why this step exists

ChartInk scans only tell us a stock triggered a pattern. They don't provide the actual OHLCV data or indicator values needed for scoring and level calculation. This step bridges that gap.

Failure Handling

Candidates that fail enrichment (API error, delisted stock, data unavailable) are silently dropped. Zero enriched -> saves empty doc, exits.

Step 4: Score Candidates

What it does

Applies a **composite scoring model** to rank candidates. Each candidate receives a score out of 100 base points, plus bonus points from confluence, scan priority, regime alignment, and weekly trend.

Base Scoring Components (100 pts max)

Component	Max Points	Logic
Close in Range (CIR)	25	How close the stock closed to its high (LONG) or low (SHORT)
Volume Ratio	25	Current volume vs average: >3x=25, >2x=20, >1.5x=15, >1.2x=10, else=5
RSI Positioning	20	LONG: 55-65 sweet spot=20; SHORT: 35-45 sweet spot=20
ATR Tradability	15	ATR%: >2.5%=15, >2.0%=10, >1.5%=5, else=0
Candle Pattern	15	Engulfing=15, Hammer=12, Directional candle=10, else=5

EMA20 Extension Filter (Pre-score gate)

- If a stock is **chasing** (LONG and >3% above EMA20, or SHORT and >3% below), it is **skipped entirely**
- If chasing between 2-3%, it receives a **-15 penalty**
- 52W scans are exempt (they are inherently extended)

Minimum Score Gate

Score must be >= 60 (MIN_SCORE) to proceed. Below this -> candidate dropped.

Bonus Points (Applied after MIN_SCORE check)

Bonus	Points	Condition
Confluence	+varies	Cluster detection across Daily / 1H / 4H pivots
Scan Priority	+0 to +10	STRONG regimes only (see table below)
Regime Alignment	+5	Trade direction matches regime
Weekly Trend Aligned	+5	LONG + weekly bullish, or SHORT + weekly bearish
Weekly Trend Contra	-10	LONG + weekly bearish, or SHORT + weekly bullish

Scan Priority Bonus (STRONG regimes only)

Scan Type	STRONG_BULL	STRONG_BEAR
breakout / breakdown	+10	+10
52W high / low	+8	+8
bull flag / bear flag	+6	+6
volume shocker	+5	+5
pullback / failed res	+3	+3
compression	+2	+2
NR7	+1	+1
inside day	+0	+0

Counter-Regime Warning

Trades that go against the regime (e.g., LONG in STRONG_BEAR) receive a **regime warning** attached to the pick. This doesn't reject the trade but flags it for position sizing reduction later.

Output

Sorted descending by rank_score . Zero scored -> saves empty doc, exits.

Step 5: Levels Engine (Entry, Stop, Target, R:R)

What it does

Every scored candidate passes through two gates. If both pass, the candidate gets precise entry, stop-loss, target, and risk-reward levels attached.

Gate 1: 1H Structural Conflict Check

- **LONG:** Rejected if a resistance zone exists within 2% above the Previous Day High (entry would slam into overhead resistance)
- **SHORT:** Rejected if a support zone exists within 2% below the Previous Day Low (entry would bounce off support)

Gate 2: calculateLevels() — Scan-Archetype-Specific

Each scan type has its own entry, stop, and target logic.

Entry Levels

Scan Type	Entry Logic
breakout_setup	Resistance level + 0.15 x ATR
pullback_at_support	EMA20 zone (4-rule decision logic)
compression / NR7 / inside (bull)	Pattern high + 0.15 x ATR
breakdown_setup	Support level - 0.15 x ATR
failed_at_resistance	EMA20 zone (4-rule decision logic)
compression / NR7 / inside (bear)	Pattern low - 0.15 x ATR

Stop-Loss Levels (All use 0.1 x ATR buffer)

Scan Type	Stop Logic
breakout_setup	Resistance level - 0.1 x ATR
pullback_at_support	Swing low - 0.1 x ATR
compression / NR7 / inside (bull)	Pattern low - 0.1 x ATR
breakdown_setup	Support level + 0.1 x ATR
failed_at_resistance	Swing high + 0.1 x ATR
compression / NR7 / inside (bear)	Pattern high + 0.1 x ATR

All stops use **ATR-based buffers** (not percentage-based). This adapts to each stock's volatility — a stock with ATR of Rs 50 gets a Rs 5 buffer, while a stock with ATR of Rs 5 gets a Rs 0.50 buffer.

Target Levels — Structural Pivot Ladder

Targets are **NOT** a simple `entry + risk x R:R` formula. The engine uses a **structural pivot ladder** that prioritizes institutional profit-taking levels:

Bullish targets cascade through:

Daily R1 -> Daily R2 -> Weekly R1 -> Weekly R2 -> 52W High -> ATR Extension -> REJECT

Bearish targets cascade through:

Daily S1 -> Daily S2 -> Weekly S1 -> Weekly S2 -> 20-Day Low -> ATR Extension -> REJECT

Each level is validated against the **minimum R:R requirement**. If a pivot is too close (fails R:R), the engine cascades to the next level. If ALL levels fail R:R, the trade is **rejected** rather than forced.

Scan-specific ladder priority:

Scan Type	Intraday Target Ladder	Swing Target Ladder
Breakout	Daily R1 -> R2 -> 3% fallback	Weekly R1 -> R2 -> 52W High
Pullback	1H Swings -> Daily R1/R2	Daily R1 -> R2 -> Weekly R1 -> R2 -> 52W High
Compression (bull)	1H Swings -> Daily R1/R2	Weekly R1 -> R2 -> 52W High
52W High	Daily R1/R2 -> 2.5x ATR	Weekly R1 -> R2 -> ATR Extension (2.5x/4.0x)
Breakdown	Daily S1 -> S2 -> 3% fallback	Weekly S1 -> S2 -> 20D Low -> ATR Extension
Failed at Resistance	1H Swings -> Daily S1/S2	Weekly S1 -> S2 -> 20D Low
Compression (bear)	1H Swings -> Daily S1/S2	Weekly S1 -> S2 -> 20D Low
52W Low	Daily S1/S2 -> 2.5x ATR	ATR Extension (2x/3x)

Partial Booking (Target 1 — 50% exit)

- **Intraday:** Daily Pivot -> midpoint between entry and main target
- **Swing:** Weekly R1 -> Daily R1 -> midpoint

Target 1 is constrained to be between entry + 2% buffer and main target - 5% buffer.

Output

allViable[] — candidates with full levels (entry, stop, target, target1, risk_pct, risk_reward, mode, target_basis) attached. R:R below minimum -> rejected.

Step 5.5: Global Market Intel (AI Web Search)

What it does

Sends the viable candidate symbols to an AI model (GPT or Claude) with **web search enabled** to fetch real-time global market intelligence. This runs at ~8:40 AM IST, providing intelligence that is current to the moment of decision-making.

Provider Options

Provider	Model	Cost
OpenAI	GPT-4.1	~Rs 55/month
Claude	Sonnet 4	~Rs 85/month

Configurable via `INTEL_PROVIDER` environment variable.

What it fetches

1. **US markets overnight** — S&P 500, Nasdaq, Dow close + Indian sector impact
2. **Asian markets** — Nikkei, Hang Seng + sentiment impact
3. **Dollar/Rupee** — DXY strength -> FII flow direction
4. **Crude oil** — Price direction -> Indian impact (import dependent economy)
5. **FII/DII flows** — Institutional buying/selling from previous session
6. **Major events** — RBI policy, budget, elections, global crises, tariffs
7. **Sector outlook** — Which NSE sectors bullish/bearish and WHY
8. **Stock-specific news** — For the exact viable symbols (earnings, SEBI actions, M&A)

STAY_OUT Gate

If the AI returns `trading_recommendation: "STAY_OUT"` or `risk_level: "EXTREME"`, the **pipeline halts entirely**. No trades that day. This covers black swan events, budget day, RBI policy announcements.

Direction Filter

- `AVOID_SHORTS` -> removes all SHORT candidates from viable pool
- `AVOID_LONGS` -> removes all LONG candidates

Score Adjustments

Sector sentiment:

Condition	Adjustment
Sector BULLISH + trade LONG	+5
Sector BEARISH + trade SHORT	+5
Sector BEARISH + trade LONG	-5
Sector BULLISH + trade SHORT	-5

Stock-specific news (HIGH impact only):

Condition	Adjustment
Aligned sentiment	+8
Opposing sentiment	-12

Re-sort

After all adjustments, viable candidates are **re-sorted** by adjusted `rank_score` descending.

SGX Data Reuse

Uses pre-fetched SGX data from Step 1 (passed via `prefetchedSGXData` parameter) to avoid a duplicate scrape.

Cache

2-hour TTL keyed by IST date. Prevents re-fetching if the pipeline retries.

Step 6: Select Diverse Picks

What it does

Selects the final trade picks from the viable pool, enforcing **sector diversity** (no correlated positions) and **scan-type diversity** (variety of setups).

Max Picks Per Regime

```
maxPicksToday = min(regime.maxTrades, MAX_DAILY_PICKS)
```

Regime	maxTrades	With MAX_DAILY_PICKS = 3
STRONG_BULL / STRONG_BEAR	3	3
WEAK_BULL / WEAK_BEAR	2	2
NEUTRAL	1	1
CONFLICT	0	Halted at Step 1

Three-Step Selection (Sector Cap First)

STEP 1 — Sector cap on full pool: Keep only the highest-scored stock per sector. If ONGC (score 88) and RELIANCE (score 82) are both ENERGY, ONGC survives and RELIANCE is dropped. UNKNOWN sector stocks are allowed through (multiple unknowns can coexist). The pool is now sector-clean.

STEP 2 — Round 1 (one per scan type): Groups the clean pool by `scan_type`. Sorts groups by their best candidate's score. Picks the top candidate from each type until slots are filled. This ensures you don't get 3 breakout picks from different stocks.

STEP 3 — Round 2 (fill by score): If slots remain after Round 1, fills from the remaining clean pool by `rank_score` regardless of scan type.

Why Sector Cap First

Previous approach (cap last) created a replacement problem:

Round 1 -> Round 2 -> Sector cap removes a pick -> now need replacement logic

Current approach (cap first) eliminates it:

Sector cap on ALL viable -> Round 1 -> Round 2 (clean pool, no replacements needed)

Example

STRONG_BULL regime, 3 slots, 6 viable candidates:

Pool after sector cap: RELIANCE (breakout, 82), TCS (compression, 76), HDFCBANK (pullback, 71), MARUTI (NR7, 65)

Round 1: picks RELIANCE from breakout (82), TCS from compression (76), HDFCBANK from pullback (71) -> 3 slots filled

Result: 3 picks, all different scan types, all different sectors.

Step 6b: AI Insights Generation (Non-fatal)

What it does

Generates a short AI-written technical insight for each selected pick using **Claude Sonnet**. The insight explains WHY the trade setup is compelling based on the technical data.

Input to AI

For each pick:

- Symbol, direction, scan type (human-readable label)
- OHLCV (open, high, low, close, prev_close, volume_ratio)
- RSI, candle pattern
- Levels (entry, stop, target, R:R)
- Market regime

System Prompt

"Ultra-brief Indian equity technical analyst. For each stock, write exactly 1-2 sentences explaining WHY this is a good intraday trade candidate based on the technical data. Focus on the setup (candle pattern, volume confirmation, key levels). Be specific with numbers."

Example Output

"Bullish engulfing at Rs 3,850 with 2.1x volume confirms breakout above resistance. RSI 62 with room to run — entry Rs 3,865, tight stop at Rs 3,830 gives 1:2.5 R:R."

Non-fatal Design

This step **never blocks the pipeline**. Three failure modes, all handled gracefully:

1. No `ANTHROPIC_API_KEY` -> skips, picks proceed with `ai_insight: null`
2. Response not parseable as JSON -> skips
3. API error (timeout, rate limit) -> catches, logs, picks proceed unchanged

Output

Each pick gets `ai_insight` (string or null) and `ai_generated` (boolean) fields attached.

Step 7: Save to DB

What it does

Saves everything into a single **DailyPick** MongoDB document, upserted by `trading_date` (one document per trading day).

Document Contents

Field	Description
<code>trading_date</code>	The date these picks will be traded
<code>scan_date</code>	The date the scans were based on (usually previous day's candle)
<code>market_context</code>	Full regime data, SGX data, global intel summary
<code>picks[]</code>	Final picks with levels, scores, AI insights, trade status
<code>summary</code>	Counts: <code>total_candidates</code> , <code>bullish_count</code> , <code>bearish_count</code> , <code>selected</code>
<code>candidates_review[]</code>	Every candidate's journey: <code>viable</code> , <code>rejected</code> (with reason), <code>selected</code>
<code>global_intel</code>	Full intel snapshot: mood, risk, sectors, stock news, events

Date Logic

- **Before 3 PM IST** (scheduled 8:40 AM run): `scan_date` = yesterday, `trading_date` = today
 - **After 3 PM IST** (manual evening run): `scan_date` = today, `trading_date` = next trading day
-

Step 8: Send Notification

What it does

Sends a **push notification** to the admin via Firebase Cloud Messaging with the day's picks summary.

Notification Content

With picks: "Daily Picks: 2 BUY + 1 SELL" with body listing symbols and entry prices.

Special cases:

- **CONFLICT** regime: "CONFLICT — Sitting Out: Structure bearish but SGX bullish"
 - **STAY_OUT** from intel: "Trading Halted: [reason from intel]"
 - **No picks:** "No picks today" with count of candidates scanned
 - **STRONG_BEAR:** Flags the bearish regime in the notification
-

Pipeline Exit Points

The pipeline has multiple early exit points. At each, it saves whatever data is available to the DB and sends a notification:

Exit Point	Condition	DB Save	Notification
Step 1	CONFLICT regime	Empty	Yes
Step 1	Regime/SGX fetch failure	None	Error alert
Step 2	Zero candidates from scans	Empty	Yes
Step 2.5	All removed by earnings	Empty	Yes
Step 3	All failed enrichment	Empty	Yes
Step 4	Zero above MIN_SCORE	Empty	Yes
Step 5.5	Global intel fetch failure	Empty	Error alert
Step 5.5	STAY_OUT / EXTREME risk	Empty	Yes (halted)
Step 5.5	Direction filter removes all	Empty	Yes
Step 6	All rejected by levels engine	Empty	Yes

Constants Reference

Constant	Value	Description
MAX_DAILY_PICKS	3	Hard cap on picks per day
MIN_SCORE	60	Minimum composite score to proceed
EARNINGS_LOOKAHEAD_DAYS	4	Days ahead to check for earnings events
INTEL_SECTOR_ALIGNED	+5	Score boost for sector-aligned trades
INTEL_SECTOR_OPPOSING	-5	Score penalty for sector-opposing trades
INTEL_STOCK_NEWS_ALIGNED	+8	Score boost for aligned stock-specific news
INTEL_STOCK_NEWS_OPPOSING	-12	Score penalty for opposing stock-specific news
ATR buffer (stops)	0.1x	Universal stop-loss buffer in ATR terms
ATR buffer (entries)	0.15x	Universal entry buffer in ATR terms
EMA20 chase skip threshold	3%	Skip candidates >3% extended from EMA20
EMA20 chase penalty range	2-3%	-15 penalty for candidates 2-3% extended
INTEL_CACHE_TTL	2h	Global intel cache expiry
EARNINGS_CACHE_TTL	4h	Earnings data cache expiry
